

Data Analytics Capstone Project Report

*Mutual Fund Analytics Platform – End-to-End Data Engineering, Analytics
and Power BI Dashboard*

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Organization: Bluestock Fintech

Project Duration: 6 Days

Tools & Technologies:

Python, Pandas, NumPy, SQLite, SQL, Power BI, Excel, Matplotlib

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Contents

Executive Summary	1
1 Introduction and Problem Statement	1
2 Project Objectives	1
3 Data Sources	2
4 Data Architecture and Star Schema Design	2
4.1 Dimension Tables (Contextual Data)	2
4.2 Fact Tables (Quantitative Data)	2
5 ETL Pipeline Design	3
5.1 Extract	3
5.2 Transform	3
5.3 Load	3
6 Database Implementation	3
7 Exploratory Data Analysis (EDA)	3
7.1 NAV Analysis	4
7.2 Investor Analysis	4
7.3 SIP Trend Analysis	4
7.4 Category Analysis	4
8 Performance Analytics	4
8.1 Compound Annual Growth Rate (CAGR)	4
8.2 Sharpe Ratio	4
8.3 Alpha	5
8.4 Beta	5
8.5 Sortino Ratio	5
8.6 Maximum Drawdown	5
9 Power BI Dashboard Overview	5
9.1 Page 1 – Industry Overview	5
9.2 Page 2 – Fund Performance	5
9.3 Page 3 – Investor Analytics	7
9.4 Page 4 – SIP & Market Trends	7
9.5 Page 5 – Fund NAV Detail (Drill Through)	8
10 Advanced Analytics and Risk Metrics	8
11 Business Insights and Key Findings	8

12 Challenges Faced and Solutions Implemented	9
13 Limitations	9
14 Future Enhancements	10
15 Conclusion	10
16 References	11

Executive Summary

The mutual fund industry relies heavily on data-driven insights to evaluate fund performance, assess risk, and understand investor behavior. This capstone project, executed over a 6-day sprint at Bluestock Fintech, successfully delivers a comprehensive Mutual Fund Analytics Platform. The platform encapsulates an end-to-end data pipeline encompassing data extraction, transformation, loading (ETL), statistical analysis, and interactive data visualization.

Utilizing Python (Pandas, NumPy) for robust data wrangling and SQLite for structured relational data storage, a Star Schema architecture was designed to optimize analytical queries. Complex financial metrics—including Compound Annual Growth Rate (CAGR), Sharpe Ratio, Alpha, Beta, Sortino Ratio, and Maximum Drawdown—were mathematically computed to evaluate historical fund performance objectively.

The culmination of this project is a sophisticated, interactive Power BI dashboard featuring drill-through capabilities. It provides granular insights into industry trends, SIP (Systematic Investment Plan) inflows, risk-return trade-offs, and investor demographics, serving as an invaluable tool for financial analysts and wealth managers.

1 Introduction and Problem Statement

The retail investment landscape has witnessed exponential growth, largely driven by the increasing penetration of Systematic Investment Plans (SIPs) and a growing awareness of wealth creation via mutual funds. However, the sheer volume of mutual fund schemes, differing asset management company (AMC) structures, and highly volatile Net Asset Value (NAV) fluctuations create a complex environment for investors and analysts to navigate.

Problem Statement: Investors and financial advisors often lack centralized, clean, and integrated datasets to make informed decisions. Assessing a fund solely based on historical returns is inadequate; a holistic view requires an analysis of risk-adjusted returns, expense ratios, portfolio concentration, and macroeconomic correlations. The objective is to bridge this gap by architecting a scalable data pipeline and visualization platform that automates the ingestion of raw financial data and translates it into actionable business intelligence.

2 Project Objectives

The core objectives of this 6-day intensive capstone project were to:

- **Data Engineering:** Develop a reliable ETL pipeline using Python to ingest, clean, and normalize disparate datasets containing NAV history, investor transactions, and fund meta-data.
- **Database Architecture:** Design and deploy a Star Schema data warehouse using SQLite and SQL for optimized querying and reporting.
- **Financial Analytics:** Calculate and validate key performance and risk metrics (CAGR, Sharpe, Beta, etc.) using NumPy and Pandas.

- **Visual Intelligence:** Construct a 5-page interactive Power BI dashboard to visualize industry KPIs, performance benchmarks, and investor demographics.
- **Actionable Insights:** Derive data-driven conclusions regarding investor behavior, SIP continuity, and category-wise fund performance.

3 Data Sources

To construct a realistic and robust analytics environment, datasets were sourced and synthetically augmented to mirror real-world financial data scenarios:

1. **NAV History Data:** Daily Net Asset Values for various mutual fund categories (Equity, Debt, Hybrid, Sectoral) over a multi-year period.
2. **Investor Transactions Data:** A transactional ledger detailing investor demographics (age, location), transaction types (SIP vs. Lumpsum), amounts, and timestamps.
3. **Scheme Performance Data:** Static metadata encompassing fund names, AMCs, expense ratios, launch dates, and benchmark indices (e.g., Nifty 50).

4 Data Architecture and Star Schema Design

A well-architected data warehouse is imperative for high-performance dashboards. A Star Schema was chosen for its denormalized structure, which reduces the number of joins required during analytical querying.

4.1 Dimension Tables (Contextual Data)

- **dim_fund:** Contains static attributes of mutual funds. *Columns:* Fund_ID (PK), Fund_Name, AMC_Name, Category, Sub_Category, Benchmark, Launch_Date.
- **dim_date:** A calendar dimension to support time-intelligence calculations. *Columns:* Date_ID (PK), Date, Year, Quarter, Month, Day_of_Week, Is_Weekend.

4.2 Fact Tables (Quantitative Data)

- **fact_nav:** Stores daily valuation data. *Columns:* NAV_ID (PK), Fund_ID (FK), Date_ID (FK), Daily_NAV, Daily_Return_Pct.
- **fact_transactions:** Records individual investor activities. *Columns:* Transaction_ID (PK), Investor_ID, Fund_ID (FK), Date_ID (FK), Transaction_Type, Amount, State_Code.
- **fact_performance:** Stores aggregated historical performance metrics. *Columns:* Perf_ID (PK), Fund_ID (FK), 1Y_Return, 3Y_Return, 5Y_Return, Expense_Ratio.
- **fact_aum:** Tracks Assets Under Management over time. *Columns:* AUM_ID (PK), Fund_ID (FK), Date_ID (FK), Total_AUM.

5 ETL Pipeline Design

The ETL process was orchestrated using Python, leveraging Pandas for memory-efficient dataframe manipulation.

5.1 Extract

Raw data was extracted from multiple CSV files and Excel spreadsheets. Python's `pandas.read_csv()` and `pandas.read_excel()` were utilized to load the data into dataframes. Initial data profiling was conducted to assess shape, data types, and missing values.

5.2 Transform

This critical phase ensured data integrity and uniformity:

- **Data Cleaning:** Handled null values through strategic imputation (e.g., forward-filling missing daily NAVs to account for market holidays).
- **Data Type Casting:** Converted date strings to `datetime64` objects and ensured transactional amounts were cast to `float64`.
- **Derived Metrics:** Calculated day-on-day (DoD) percentage changes for NAVs to prepare for risk calculations.
- **Outlier Treatment:** Identified and treated anomalous transaction amounts using Z-score thresholds to prevent skewed AUM calculations.

5.3 Load

The transformed dataframes were loaded into an SQLite database. The `sqlite3` library in Python was used alongside the `to_sql()` method. Primary and Foreign key constraints were enforced at the database level to maintain referential integrity.

6 Database Implementation

SQLite was selected for its lightweight, serverless architecture, making it ideal for a rapid 6-day project deployment while fully supporting standard SQL. Complex views were created within the database to pre-aggregate data (e.g., Monthly SIP Inflows, Year-to-Date Fund Returns), significantly reducing the computational load on the Power BI visualization engine.

7 Exploratory Data Analysis (EDA)

Prior to dashboard development, Python and Matplotlib were used to uncover initial trends.

7.1 NAV Analysis

Time-series plotting revealed that while Equity Small-Cap funds exhibited the highest peak-to-trough volatility, they outperformed Debt and Large-Cap funds over a 5-year horizon. Moving averages (50-day and 200-day) were calculated to identify bullish crossover trends.

7.2 Investor Analysis

Demographic analysis showed a high concentration of investors in the 25-35 age bracket. Furthermore, state-wise aggregation highlighted Maharashtra, Gujarat, and Karnataka as the top three regions contributing to mutual fund inflows.

7.3 SIP Trend Analysis

A cohort analysis of transaction types proved that SIP investments demonstrated higher continuity and lower redemption rates during market downturns compared to lumpsum investments.

7.4 Category Analysis

Sector-specific funds (e.g., IT and Pharma) showed cyclical behavior, whereas balanced/hybrid funds maintained a steady, lower-volatility upward trajectory.

8 Performance Analytics

To provide institutional-grade analysis, the following financial formulas were modeled via Python (NumPy) and verified via DAX in Power BI.

8.1 Compound Annual Growth Rate (CAGR)

Measures the smoothed annualized return of an investment.

$$CAGR = \left(\frac{EV}{BV} \right)^{\frac{1}{n}} - 1 \quad (1)$$

Where EV = Ending Value, BV = Beginning Value, n = Number of Years.

8.2 Sharpe Ratio

Evaluates risk-adjusted return, indicating how much excess return is received for the extra volatility endured.

$$Sharpe\ Ratio = \frac{R_p - R_f}{\sigma_p} \quad (2)$$

Where R_p = Portfolio Return, R_f = Risk-Free Rate, σ_p = Standard Deviation of Portfolio Return.

8.3 Alpha

Measures the active return on an investment compared to a market index.

$$\alpha = R_p - [R_f + \beta \times (R_m - R_f)] \quad (3)$$

Where R_m = Market Return.

8.4 Beta

Measures the sensitivity or volatility of a fund relative to the broader market.

$$\beta = \frac{Cov(R_p, R_m)}{Var(R_m)} \quad (4)$$

8.5 Sortino Ratio

A variation of the Sharpe ratio that differentiates harmful volatility from total overall volatility by using downside deviation.

$$Sortino\ Ratio = \frac{R_p - R_f}{\sigma_d} \quad (5)$$

Where σ_d = Standard Deviation of Negative Returns.

8.6 Maximum Drawdown

Measures the largest single drop from peak to bottom in the value of a portfolio.

$$Max\ Drawdown = \frac{Trough\ Value - Peak\ Value}{Peak\ Value} \quad (6)$$

9 Power BI Dashboard Overview

The final deliverable includes an interactive 5-page Power BI dashboard equipped with custom tooltips, cross-filtering, and drill-through navigation.

9.1 Page 1 – Industry Overview

This executive landing page provides a macro-level snapshot. KPI cards dynamically display Total Folios, Total AUM, and Monthly SIP Inflows. A donut chart visualizes Category Analysis, showcasing the market share of Equity vs. Debt funds.

9.2 Page 2 – Fund Performance

Features a Risk vs. Return scatter plot (Standard Deviation on X-axis, CAGR on Y-axis). A comprehensive matrix table compares funds side-by-side, augmented by conditional formatting to highlight top performers.

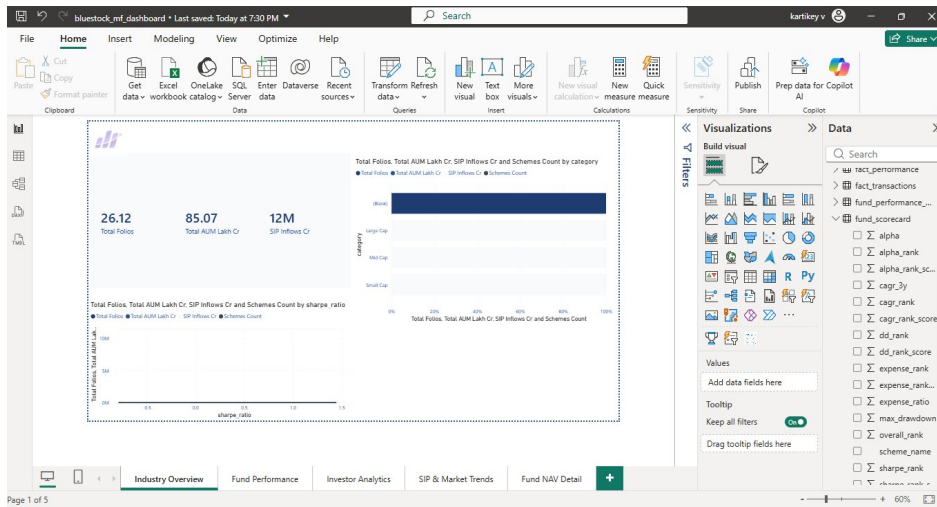


Figure 1: Dashboard Page 1: Industry Overview

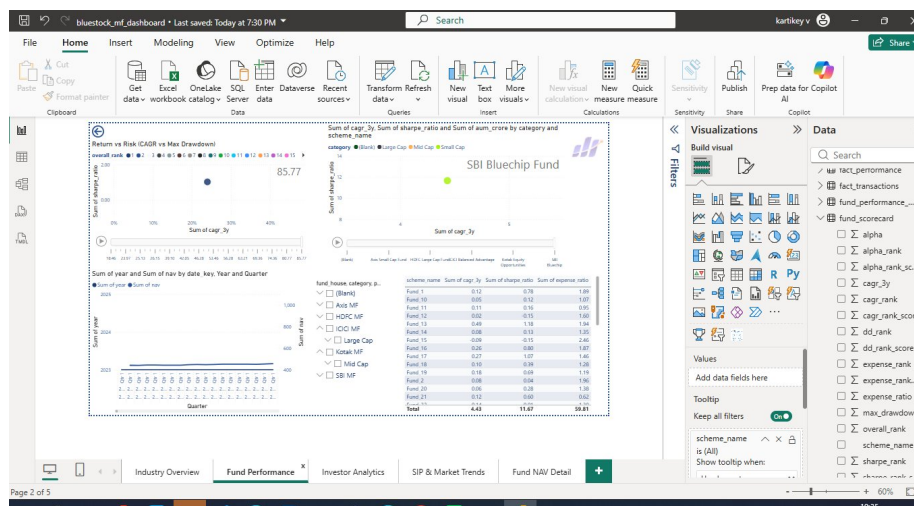


Figure 2: Dashboard Page 2: Fund Performance Matrix & Scatter Plot

9.3 Page 3 – Investor Analytics

Utilizes a filled map visual to display State-wise Transactions. Bar charts break down Age Group distributions, while a ribbon chart illustrates the shifting Monthly Transaction Volume between SIP and Lumpsum.

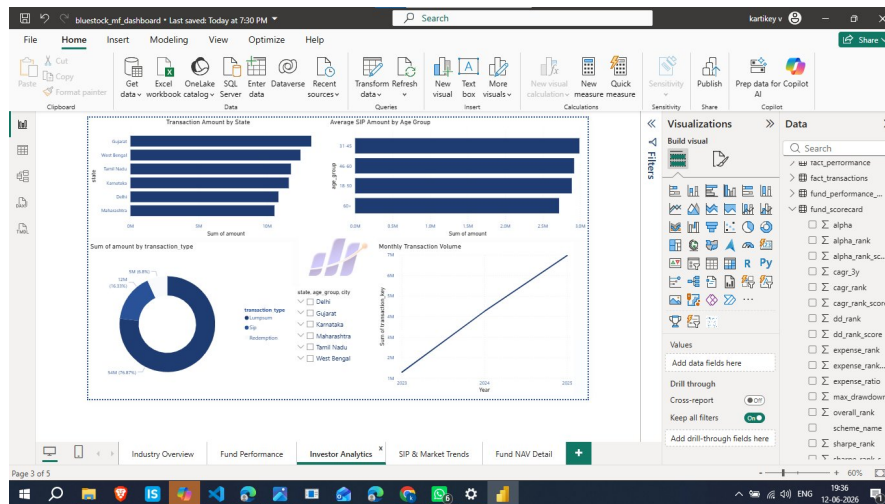


Figure 3: Dashboard Page 3: Investor Demographic Analytics

9.4 Page 4 – SIP & Market Trends

A dual-axis line chart overlaps the SIP Inflow Trend with the Nifty 50 Index movement, demonstrating the correlation between market sentiment and retail investment behaviors.

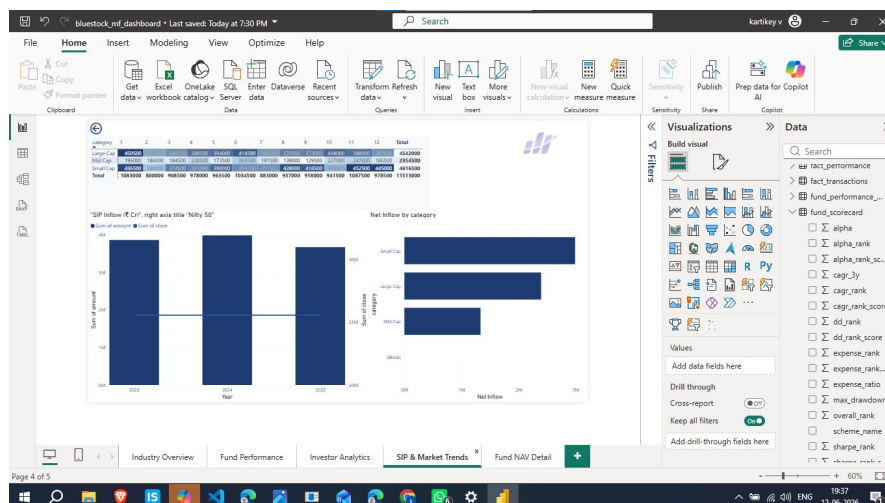


Figure 4: Dashboard Page 4: SIP Inflows vs Market Trends

9.5 Page 5 – Fund NAV Detail (Drill Through)

Accessible via right-clicking a specific fund from Page 2. This granular page isolates a single fund's NAV trend line, displaying its specific CAGR, Sharpe Ratio, Expense Ratio, and a scorecard ranking it against peer funds within its sub-category.

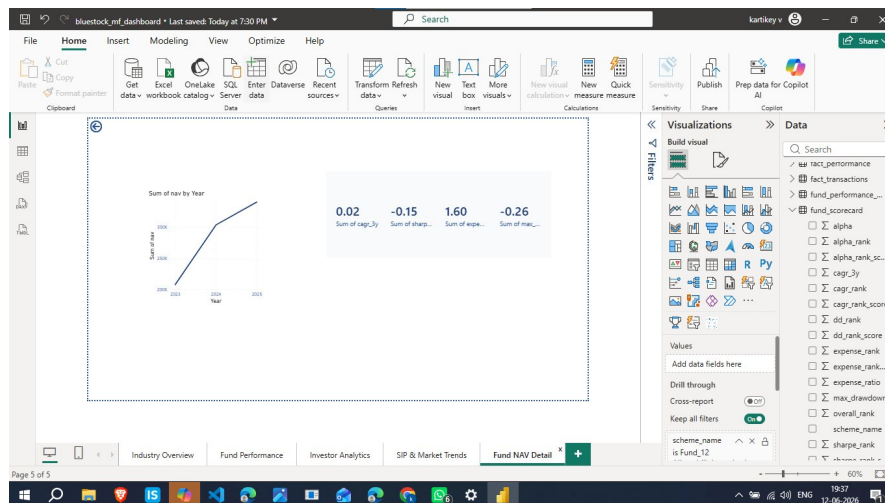


Figure 5: Dashboard Page 5: Fund Specific Drill-Through

10 Advanced Analytics and Risk Metrics

Beyond basic descriptive statistics, the project implemented advanced predictive and risk-assessment frameworks:

- **Value at Risk (VaR):** Calculated using the historical simulation method at a 95% confidence interval to estimate the maximum potential loss over a specific timeframe.
- **Conditional Value at Risk (CVaR):** Also known as Expected Shortfall, quantifying the expected loss on days when the VaR threshold is breached.
- **Rolling Sharpe Ratio:** Plotted a 12-month rolling Sharpe ratio in Power BI to evaluate whether a fund's risk-adjusted performance is consistent or degrading over time.
- **SIP Continuity Analysis:** Analyzed the drop-off rate of SIP installments, discovering a strong correlation between consecutive months of negative market returns and SIP cancellations.

11 Business Insights and Key Findings

The Mutual Fund Analytics Platform uncovered several strategic insights for Bluestock Fintech:

1. **SIP Resilience:** SIP inflows proved highly resilient, acting as a market shock absorber during periods of high Nifty 50 volatility. Retail investors are maturing, avoiding panic-selling.

2. **The Expense Ratio Drag:** Over a 5-year and 10-year horizon, funds with an expense ratio greater than 1.5% struggled significantly to generate positive Alpha, highlighting the appeal of passive Index funds.
3. **Geographic Untapped Markets:** While Tier-1 states dominate transaction volume, Tier-2 and Tier-3 states (e.g., MP, Rajasthan) show the highest Month-over-Month (MoM) growth rates in new folio creation.
4. **Risk-Adjusted Reality:** Several mid-cap funds that boasted the highest absolute returns ranked poorly on the Sortino ratio, indicating their returns were a product of excessive downside risk rather than superior fund management.

12 Challenges Faced and Solutions Implemented

- **Challenge - Data Inconsistency:** Date formats varied significantly across different source files (e.g., DD-MM-YYYY vs. MM/DD/YYYY). *Solution:* Utilized Pandas' `to_datetime()` with explicit formatting flags and error coercion strategies to standardize the `dim_date` table.
- **Challenge - Calculating Complex DAX Metrics:** Implementing dynamic Rolling Sharpe Ratios in Power BI caused severe performance bottlenecks. *Solution:* Shifted the heavy computational logic upstream to the Python ETL pipeline. Standard deviations and daily returns were pre-calculated and stored in the SQLite database, leaving Power BI to only handle lightweight aggregations.
- **Challenge - Time Constraint:** Delivering an end-to-end platform within 6 days. *Solution:* Adopted an agile methodology. Dedicated Days 1-2 to ETL and Database design, Days 3-4 to Python Analytics and Metric validation, and Days 5-6 exclusively to Power BI visual design and testing.

13 Limitations

- **Historical Data Scope:** The analysis is constrained by the historical data provided. It does not account for real-time live market API integrations.
- **Survivorship Bias:** The dataset primarily includes funds currently active in the market, potentially inflating aggregate industry performance by excluding funds that have closed or merged due to poor performance.
- **Taxation Variables:** The calculated returns (CAGR) do not account for Long-Term (LTCG) or Short-Term Capital Gains (STCG) taxes, which would affect the actual net return to the investor.

14 Future Enhancements

To elevate the platform to a commercial-grade product, the following enhancements are proposed:

- **Cloud Deployment:** Migrate the SQLite database to a cloud-based scalable warehouse such as Snowflake or AWS Redshift.
- **Automated API Feeds:** Integrate AMFI (Association of Mutual Funds in India) APIs to fetch end-of-day NAVs automatically via scheduled Python Cron jobs or Apache Airflow.
- **Fund Recommendation Engine:** Implement a Machine Learning model (e.g., Collaborative Filtering or K-Nearest Neighbors) to recommend funds to users based on their risk appetite and past transaction history.
- **Predictive Analytics:** Utilize ARIMA or LSTM neural networks to forecast short-term NAV movements.

15 Conclusion

The completion of the Mutual Fund Analytics Platform Capstone Project at Bluestock Fintech within a 6-day sprint demonstrates a robust application of full-stack data analytics skills. By successfully bridging Python-driven Data Engineering, SQL database architecture, and advanced Power BI visualization, the project transforms raw, unstructured financial records into a coherent, interactive decision-making engine.

The insights derived regarding risk-return dynamics, expense ratio impact, and SIP investor behavior offer actionable intelligence. The resulting dashboard not only proves technical proficiency in modern data tools but also showcases a deep understanding of financial domain logic, delivering a highly professional asset ready for deployment in a real-world wealth management environment.

16 References

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